Lecture 25: Agents, tools, and the agentic era

PSYC 51.17: Models of language and communication

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Learning objectives

By the end of this lecture, you will be able to...

1. Define what an **LLM agent** is and explain the **ReAct** reasoning-action loop
2. Describe **function calling** and the **Model Context Protocol (MCP)** — the universal standard for tool integration
3. Evaluate how **coding agents** (Claude Code, Cursor, Devin) are reshaping software development
4. Explain **computer use** and **deep research** — agents that see your screen and browse the web autonomously
5. Assess the **safety implications** of giving LLMs increasing autonomy to act in the world

From chatbots to agents

What is an LLM agent?

An **agent** is a system where a language model can:

1. **Reason** about a task (plan steps, reflect on progress)
2. **Act** by calling external tools (search, code execution, APIs)
3. **Observe** the results of those actions
4. **Iterate** until the task is complete

A chatbot *responds to prompts*. An agent *takes actions in the world.*

Questions to consider

When you use ChatGPT to browse the web or run Python code, you're interacting with an agent. What makes this qualitatively different from a simple question-answering system?

Why tools matter

LLMs are powerful but limited

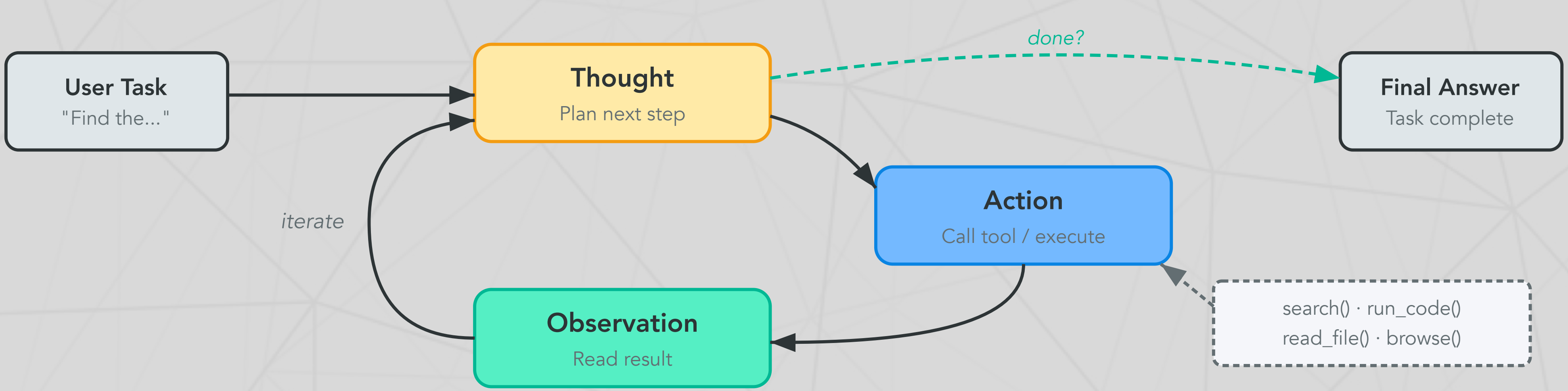
Language models trained on text alone cannot:

- **Access current information** (training data has a cutoff date)
- **Perform precise computation** (arithmetic, symbolic math)
- **Interact with external systems** (databases, APIs, file systems)
- **Verify their own claims** (no ground truth access)

Tools compensate for LLM weaknesses

By connecting an LLM to external tools, we combine the model's **language understanding and reasoning** with tools that provide **accuracy, recency, and real-world interaction**. The LLM decides *what* to do; the tools *do* it.

The ReAct framework



ReAct in action

```
Question: "What is the elevation of the city where Dartmouth is located?"

Thought 1: I need to find which city Dartmouth College is in.
Action 1:  search("Dartmouth College location")
Obs 1:     Dartmouth College is in Hanover, New Hampshire.

Thought 2: Now I need the elevation of Hanover, NH.
Action 2:  search("Hanover New Hampshire elevation")
Obs 2:     Hanover, NH has an elevation of 531 feet (162 m).

Thought 3: I have the answer.
Action 3:  finish("531 feet (162 meters)")
```

Function calling

How LLMs use tools

Function calling is a mechanism where the LLM outputs a structured JSON request to invoke an external function. The system executes the function and feeds the result back to the LLM.

The three-step dance

```
# Step 1: LLM generates a function call (not free text)
response = {"function_call": {"name": "get_weather",
    "arguments": '{"location": "Hanover, NH"}'}}
# Step 2: System executes the function
result = get_weather(location="Hanover, NH")  # → {"temp": 28, "condition": "snowy"}
# Step 3: Result fed back → LLM generates: "It's 28°F and snowy in Hanover."
```

Key insight

The LLM never *executes* code itself — it generates a **structured request** that the system dispatches. This separation of reasoning from execution is fundamental to agent safety.

Model Context Protocol (MCP)

USB-C for AI — Anthropic (November 2024)

MCP is an open protocol that standardizes how LLMs connect to external tools and data sources. Any MCP-compatible tool works with any MCP-compatible model.

Adoption explosion

Metric	Nov 2024 (launch)	Apr 2025	Feb 2026
MCP servers	~10	5,800+	Growing rapidly
MCP clients	~3	300+	All major platforms
SDK downloads/month	100K	8M	**97M**
Supported by	Anthropic	+ OpenAI, Google	+ Microsoft, all major IDEs

Why MCP won

Within 14 months, MCP became the de facto universal standard. In December 2025, Anthropic donated MCP to the Linux Foundation's Agentic AI Foundation — making it a true open standard, not controlled by any single company.

Coding agents: SWE-bench in 18 months

From 14% to 81% in less than two years

SWE-bench Verified tests whether an agent can fix real GitHub issues — read the codebase, understand the bug, write a fix, and pass the test suite.

Date	Agent	SWE-bench Verified
Mar 2024	Devin (first "AI software engineer")	13.9%
Oct 2024	Claude 3.5 Sonnet	49.0%
Feb 2025	Claude 3.7 Sonnet	62.3%
Apr 2025	GPT-5	74.9%
Nov 2025	**Claude Opus 4.5**	**80.9%** (first to break 80%)

What this means

Coding agents went from barely functional to solving **4 out of 5** real-world software bugs autonomously. Claude Code — Anthropic's terminal-based coding agent — reached **$1 billion** in annualized revenue within 6 months of launch.

How coding agents work

The agent loop in practice

Nearly all coding agents follow the same core loop:

1. **Read** the codebase (file system access)
2. **Plan** a sequence of changes
3. **Edit** files and write new code
4. **Run** tests and commands (shell access)
5. **Observe** results, fix errors
6. **Iterate** until the task passes or a limit is reached

The ecosystem

Tool	Form factor	Key strength
Claude Code	Terminal CLI	Works on remote servers, CI/CD, any language
Cursor	IDE	Deep editor integration, multi-file edits
Devin	Autonomous agent	End-to-end task completion, acquired Windsurf IDE
GitHub Copilot Workspace	GitHub-native	Plans changes across entire repos

Computer use: agents that see your screen

Anthropic (October 2024 — present)

Computer use gives Claude the ability to see screenshots, move the mouse, click buttons, and type — interacting with *any* software, not just tools with APIs.

Three approaches to computer control

System	Developer	How it works	OSWorld score
Computer Use	Anthropic	Screenshots + keyboard/mouse control	**72.5%**
Operator (CUA)	OpenAI	Virtual browser environment	38.1%
Project Mariner	Google	Cloud VM, up to 10 parallel tasks	—

The trajectory

Claude's computer use score on OSWorld improved **3.3×** in 16 months: 22% (Oct 2024) → 72.5% (Feb 2026). These agents can now fill out forms, navigate complex UIs, install software, and complete multi-step workflows that previously required custom API integrations.

Deep research agents

Autonomous multi-hour research (all launched February 2025)

All three major AI companies shipped autonomous research agents within an 11-day window:

Agent	Developer	How it works
Deep Research	OpenAI	o3 variant + web browsing; 5–30 min reports
Deep Research	Google	Gemini 3 Pro + Google Search; 100+ pages per query
Deep Research	Perplexity	Parallelized ingestion; strong source tracing

The capability jump

OpenAI's Deep Research scored **26% on Humanity's Last Exam** — when standard models scored 1–5%. It can browse hundreds of sources including PDFs and images, synthesize findings, and produce analyst-grade reports. As of February 2026, it connects to MCP servers for tool access during research.

Limitation

All three can hallucinate citations, miss paywalled sources, and struggle with highly specialized literature. Independent verification remains essential.

Multi-agent systems

Multiple LLMs collaborating on complex tasks

Multi-agent systems use multiple LLM instances — often with different roles — to tackle problems too complex for a single agent:

Common architectures

Pattern	How it works	Example
Supervisor + Workers	Orchestrator delegates to specialists	Manager assigns code review to specialist agents
Peer-to-peer	Agents communicate directly	Two agents debate a solution
Swarm	Many parallel agents on shared task	Kimi K2.5 uses parallel reasoning
Pipeline	Sequential handoff between specialists	Research agent → Analysis agent → Writing agent

Frameworks

Leading frameworks: **LangGraph** (finite state machines), **CrewAI** (role-based teams), **AutoGen** (Microsoft, research-focused), **MetaGPT** (ICLR 2024 oral — "AI Software Company").

Agent memory

How agents remember across long tasks

LLM context windows are finite (128K–1M tokens). Long-running agents need additional memory:

Memory type	Implementation	Use case
Short-term	Conversation history in context	Recent steps and observations
Working memory	Scratchpad / notepad tool	Intermediate results, running totals
Long-term	Vector database (Lecture 17)	Past experiences, learned procedures
Episodic	Structured logs	What worked/failed in previous runs

The memory bottleneck

Memory management is one of the hardest problems in agent design. Too little context and the agent forgets its plan. Too much and it becomes slow and confused. The trend: **hierarchical memory** — compress old context rather than discarding it. Long context windows (Gemini at 1M tokens) help but don't fully solve this.

Agent safety: a growing concern

The International AI Safety Report (February 2026)

The 2026 International AI Safety Report — a consensus document from researchers worldwide — found:

- AI agents can identify **77% of vulnerabilities** in real software in controlled competitions
- Criminal groups and state actors are **actively using** general-purpose AI in operations
- Multiple companies could not rule out bioweapons uplift before deploying; added heightened safeguards
- Technical safeguards are improving but not complete

Agent-specific attack surface

Attack	How it works	Mitigation
Prompt injection	Malicious web content hijacks agent mid-task	Input sanitization, sandboxing
Tool misuse	Agent with write permissions induced to take harmful actions	Minimal capability grants, human approval
Malicious MCP servers	Supply-chain attack via compromised tool server	Server verification, audit logging
Cascading errors	One bad tool call triggers a chain of incorrect actions	Step limits, rollback mechanisms

When agents go wrong: real-world examples

Agents that delete your work (July 2025)

A developer reported that **Gemini CLI hallucinated file operations** — silently creating directories in wrong locations, then running `move *` commands that overwrote files. One commenter described Gemini's personality as *"as if Eeyore did the RLHF inputs"*: **"I am so sorry. I have made another inexcusable error."** Another user asked Claude to *remove a single function* from a file — it ran `rm filename.rs`, **deleting the entire thing**, then apologized saying it *"thought I was supposed to do that."*

"Let's try a different approach"

Users identified a terrifying pattern: when an agent gets stuck, it says *"Let's try a different approach"* — which, as one commenter put it, precedes the agent entering **"paperclip mode"** where it destroys everything rather than admitting defeat. Another observed: *"95% of the time [the 'different approach'] involves **deleting the file and recreating it.** It's mind-blowing it happens so often."*

Easy problems that LLMs get wrong (Nezhurina et al., 2024)

Nezhurina et al. (2024) tested frontier models on 30 questions **humans find trivial** (86% accuracy). GPT-4 scored **38%**, Claude 3 Opus **35%**. Gems: *"How many L's in LOLLAPALOOZA?"* — GPT-4: **"5"** (it's 4). *"Sally has 3 brothers. Each brother has 2 sisters. How many sisters does Sally have?"* — GPT-4: **"2"** (it's 1). When agents can't count letters, giving them tools to *act on* such reasoning is risky.

The agent spectrum

Low autonomy — High autonomy

Low risk — High risk

1	2	3	4	5
Tool-using	**ReAct Agent**	**Coding Agent**	**Computer Use**	**Multi-Agent**
ChatGPT+plugins	Research assistant	Claude Code, Devin	Screen control	LLMs → LLMs

The autonomy dilemma

Each step up gives agents more capability — and more potential for harm. Anthropic published a framework for measuring agent autonomy that categorizes tasks by reversibility, blast radius, and required human oversight. The principle: **match autonomy to the stakes**.

Discussion: the automation frontier

Questions that will shape your career

1. **The coding question:** Claude Code can now fix 81% of real GitHub bugs autonomously. It reached $1B in revenue in 6 months. If you're a computer science student, how does this change what you should learn? Is this different from how compilers automated assembly language — or is it different in kind?
2. **The trust problem:** Computer use agents can see your screen, click buttons, and fill out forms. Would you trust an AI to file your taxes? Book your flights? Send emails on your behalf? Where's *your* trust boundary — and why there?
3. **The MCP ecosystem:** MCP standardizes tool access so any model can use any tool. If tools are interchangeable and models are interchangeable, where does the value lie? Who benefits most from open standards vs. proprietary ecosystems?
4. **The principal-agent problem:** When you delegate a task to an AI agent, how do you verify it did what you wanted — especially when its reasoning is opaque? How is this different from delegating to a human employee?

Further reading

Further reading

Yao et al. (2023, *ICLR*) "ReAct: Synergizing Reasoning and Acting in Language Models" — The ReAct framework.

Anthropic (2024) "Model Context Protocol" — Open standard for LLM-tool integration (now Linux Foundation).

Jimenez et al. (2024, *ICLR*) "SWE-bench: Can Language Models Resolve Real-World GitHub Issues?" — The benchmark for coding agents.

International AI Safety Report (2026) "International Scientific Report on AI Safety" — Global consensus on agent risks.

Anthropic (2025) "Measuring Agent Autonomy" — Framework for categorizing agent risk levels.

Questions?

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Up next...

The reckoning: society, safety, and what comes next